

Python Basics - Study Notes

1. What is Python?

Python is a high-level, interpreted programming language. It is easy to read and is used in web development, data science, artificial intelligence, machine learning, and automation.

2. Variables

A variable is a name used to store a value. Example: `age = 20`. Python automatically determines the data type of the value.

3. Data Types

Common Python data types include `int`, `float`, `str`, `bool`, `list`, and `dict`.

4. Conditional Statements

The `if`, `elif`, and `else` statements are used to make decisions in a program.

5. Loops

Loops repeat a block of code. Python commonly uses `for` loops and `while` loops.

6. Functions

A function is a reusable block of code. It is defined using the `def` keyword.

7. Lists

A list stores multiple values in one variable. Example: `fruits = ['apple', 'banana', 'mango']`.

8. Why is Python useful?

Python has simple syntax, many libraries, and a large community. It is useful for beginners and professional developers.